

Transient receptor potential TRPA1 channel desensitization in sensory neurons is agonist dependent and regulated by TRPV1-directed internalization

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The pharmacological desensitization of receptors is a fundamental mechanism for regulating the activity of neuronal systems. The TRPA1 channel plays a key role in the processing of noxious information and can undergo functional desensitization by unknown mechanisms. Here we show that TRPA1 is desensitized by homologous (mustard oil; a TRPA1 agonist) and heterologous (capsaicin; a TRPV1 agonist) agonists via Ca^{2+} -independent and Ca^{2+} -dependent pathways, respectively, in sensory neurons. The pharmacological desensitization of TRPA1 by capsaicin and mustard oil is not influenced by activation of protein phosphatase 2B. However, it is regulated by phosphatidylinositol-4,5-bisphosphate depletion after capsaicin, but not mustard oil, application. Using a biosensor, we establish that capsaicin, unlike mustard oil, consistently activates phospholipase C in sensory neurons. We next demonstrate that TRPA1 desensitization is regulated by TRPV1, and it appears that mustard oil-induced TRPA1 internalization is prevented by coexpression with TRPV1 in a heterologous expression system and in sensory neurons. In conclusion, we propose novel mechanisms whereby TRPA1 activity undergoes pharmacological desensitization through multiple cellular pathways that are agonist dependent and modulated by TRPV1.

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The dynamics of nervous system plasticity depend on the balance between functional sensitization and desensitization of membrane receptors. One physiological function of certain TRP channels is to mediate the responses of cells, including sensory neurons, to chemical and physical stimuli (Clapham, 2003; Numazaki & Tominaga, 2004). The TRPA1 channel participates in the transmission of noxious information by sensory neurons as well as the development of hypersensitivity to noxious stimulations (i.e. hyperalgesia) in several pain models (Obata *et al.* 2005; Bautista *et al.* 2006; Kwan *et al.* 2006). Thus, regulation of channel activity by sensitization or desensitization could contribute to overall sensory neuron function and nociceptive processing.

It is well documented that pre-treatment with chemical irritants, such as mustard oil (MO) and capsaicin (CAP), can lead to functional desensitization of MO responses in a variety of sensory systems including the trigeminal system (Reeh *et al.* 1986; Patacchini

et al. 1990; Heyer *et al.* 1991; Brand & Jacquot, 2002; Simons *et al.* 2004; Jacquot *et al.* 2005). Functional desensitization of MO responses could be the consequence of acute (attenuation of responses during constant drug application) and/or pharmacological (diminished response after pre-treatment with MO or another drug) desensitization of the TRPA1 channel in sensory neurons. However, the molecular mechanisms underlying TRPA1 desensitization is an important question that remains without a definitive answer.

Desensitization of TRP channels is mediated by at least one of three different cellular pathways. First, in sensory neurons and heterologous expression systems, TRPV1 desensitization is due to CAP-, ATP- or cannabinoid-evoked activation of the Ca^{2+} -dependent phosphatase, calcineurin, leading to dephosphorylation and acute (Docherty *et al.* 1996; Koplas *et al.* 1997) or pharmacological desensitization (Piper & Docherty, 2000; Mohapatra & Nau, 2005; Jeske *et al.* 2006) of the TRPV1 channel. Second, in heterologous expression systems, acute desensitization of Ca^{2+} -activated TRPM4 is controlled by phosphatidylinositol 4,5-bisphosphate

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(PIP₂) depletion that is triggered by an elevation in intracellular Ca²⁺ (Zhang *et al.* 2005; Nilius *et al.* 2006) and TRPM5 (Nilius *et al.* 2005). The acute desensitization of the Ca²⁺-activated Ca²⁺ channels TRPV5 (Rohacs *et al.* 2005) and TRPV6 (Nilius *et al.* 2005) employs a similar mechanism. Further, menthol-gated Ca²⁺ influx through TRPM8 activates a Ca²⁺-sensitive phospholipase C (PLC) and subsequently results in PIP₂ hydrolysis and TRPM8 pharmacological desensitization (Rohacs *et al.* 2005). A possible role of PIP₂ depletion has also been proposed for TRPV1 desensitization in HEK cells (human embryonic kidney cell line) (Liu *et al.* 2005). The third general mechanism for TRP desensitization is via activation of Ca²⁺-dependent protein kinase C (PKC) leading to phosphorylation-based desensitization of TRPM8 (Abe *et al.* 2006) and TRPC5 (Zhu *et al.* 2005). Pharmacological assays also indicate that α and/or β_1 subtypes of conventional Ca²⁺-dependent PKC isoforms mediate menthol-induced TRPM8 desensitization (Abe *et al.* 2006). Altogether, TRP channel desensitization involves either dephosphorylation of the channels by calcineurin, regulation of channel activities by PIP₂ depletion/replenishment or phosphorylation of the channels by PKC isoforms. These cellular pathways have a common feature: a dependency upon an elevation in intracellular Ca²⁺ levels.

A main aim of the present study was to characterize the cellular pathways involved in the pharmacological desensitization of TRPA1 by CAP and MO in sensory neurons. Our data demonstrate that TRPA1 desensitization in sensory neurons is regulated either by CAP-induced Ca²⁺-dependent PIP₂ depletion or a by unique pathway involving Ca²⁺-independent MO-directed inhibition of the channel. Results presented here also imply that the magnitude of TRPA1 desensitization is influenced by TRPV1 coexpression in sensory neurons that prevents TRPA1 internalization.

Methods

Animals, primary sensory neuron culture, constructs and expression in trigeminal ganglia (TG) and Chinese hamster ovary (CHO) cells

Sprague–Dawley rats, 45–60 days old, were obtained from a commercial breeder (Charles River Laboratories, Inc., Wilmington, MA, USA, or Harlan, Indianapolis, IN, USA). B6.129S4 or B6.129S4-trpV1^{tml/jul} (TRPV1 null-mutant) mice, 40–60 days old, were obtained from The Jackson Laboratory (Bar Harbor, ME, USA). All experiments conformed to protocols approved by the University of Texas Health Science Center at San Antonio (UTHSCSA) Animal Care and Use Committee (ACUC). We followed guidelines issued by the National Institutes of Health and

the Society for Neuroscience to minimize the number of animals used and their suffering.

Mice and rats were deeply anaesthetized with isoflurane (0.3 ml in 1 l administered for 60–90 s) and killed by decapitation. The trigeminal ganglia (TG) were quickly removed from the skull and placed in ice-cold Hank's solution (Invitrogen, Carlsbad, CA, USA). TG neurons were separated by treatment in a 1 mg ml⁻¹ collagenase–dispase (Roche, Indianapolis, IN, USA) solution. Neurons were plated at low-density on poly D-lysine–laminin-coated coverslips (Clontech, Palo Alto, CA, USA). Cells were maintained in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS), 2 mM L-glutamine, 100 U ml⁻¹ penicillin, 100 μ g ml⁻¹ streptomycin and 100 ng ml⁻¹ NGF-7.02S (Harlan). The use of dispase instead of trypsin during separation of neurons and nerve growth factor (NGF) in the maintenance media was critical to achieving reproducible and consistent responses to MO application. The experiments were performed 24–36 h after plating.

Expression plasmids of TRPV1 (accession number NM031982) in pcDNA3 (Invitrogen); TRPA1 (NM177781) in pcDNA5/FRT (Invitrogen) and PHD-GFP (PLC activity bio-sensor) were used. Expression constructs were delivered into CHO cells using PolyFect (Qiagen, Valencia, CA, USA) and into TG neurons using nucleoporator Nucleofector II (Amaxa) and Rat Dorsal Root Ganglion kit (Amaxa) according to manufacturers' protocols. CHO cells and TG neurons were subjected to experimental procedures within 24–48 h after transfection.

Electrophysiology

Recordings were made in perforated patch or whole-cell voltage clamp (holding potential (V_h) of -60 mV) configurations at 22–24°C from the somata of neurons (15–40 pF) or CHO cells. Data were acquired and analysed using an Axopatch 200B amplifier and pCLAMP9.0 software (Axon Instruments, Union City, CA, USA). Recording data were filtered at 0.5–2.5 kHz and sampled at 2–10 kHz depending on current kinetics. Borosilicate pipettes (Sutter, Novato, CA, USA) were polished to resistances of 4–7 M Ω in the perforated patch pipette solution (3–4 M Ω in whole-cell pipette solution). Access resistance (R_s) was compensated (40–80%) when appropriate up to the value of 13–18 M Ω for perforated patch or 7–10 M Ω for whole-cell configurations. Data were rejected when R_s changed $> 20\%$ during recording, leak currents were > 50 pA, or input resistance was < 300 M Ω . Currents were considered positive when their amplitudes were 5-fold bigger than displayed noise (in root mean square).

Standard external solution (SES) for whole-cell and perforated patch recording contained (mM): 140 NaCl, 5 KCl, 2 CaCl₂, 1 MgCl₂, 10 D-glucose and 10 Hepes, pH 7.4. In Ca²⁺-free external solution (0 Ca-ES), 5 mM EGTA was added to buffer ambient Ca²⁺. Vehicle, which is 0.01% DMSO, was added to external solutions. To record high-voltage-activated (HVA) calcium currents (I_{Ca}), the external solution was (mM): 160 TEA-Cl, 2.5 CaCl₂, 10 D-glucose, 10 Hepes pH 7.4. The pipette solution for the perforated patch configurations consisted of (mM): 110 potassium methanesulphonate, 30 KCl, 1 MgCl₂, 10 Hepes pH 7.3 and 250 $\mu\text{g ml}^{-1}$ amphotericin B (Sigma, St Louis, MO, USA). The standard pipette solution (SIS) for the whole-cell configurations contained (mM): 140 KCl, 1 MgCl₂, 1 CaCl₂, 10 EGTA, 10 D-glucose, 10 Hepes, pH 7.3. In a set of experiments designed to suppress voltage-activated K⁺ currents, KCl was substituted with equimolar CsCl. Drugs were applied using a fast, pressure-driven and computer-controlled 8-channel system (ValveLink8; AutoMate Scientific, San Francisco, CA, USA).

Fluorescence imaging in TG neurons and CHO cells

Fluorescence imaging was basically performed as previously described (Jeske *et al.* 2006). Fluorescence was detected by a Nikon TE 2000U microscope fitted with a $\times 40/1.35$ NA Fluor objective. Data were collected and analysed with MetaFluor Software (MetaMorph, Universal Imaging Corporation, Downingtown, PA, USA). The experiments were performed in SES solution (see Electrophysiology section). The calcium-sensitive dye was the acetoxymethyl ester form of Fura-2, Fura-2 AM (2 μM ; Molecular Probes, Carlsbad, CA, USA). The net changes in Ca²⁺ influx were calculated by subtracting the basal [Ca²⁺]_i (mean value collected for 60 s prior to agonist addition) from the peak [Ca²⁺]_i value achieved after exposure to the agonists. Increases in [Ca²⁺]_i above 50 nM were considered positive. This minimal threshold criterion was established by application of 0.1% DMSO as a vehicle. Ratiometric data were converted to [Ca²⁺]_i (μM) as previously described (Jeske *et al.* 2006).

Biotinylation of cell surface proteins

CHO cells or cultured TG neurons were rinsed three times with ice-cold PBS and were incubated with 0.5 mg ml⁻¹ EZ-Link Sulfo-NHS-LC-Biotin (Pierce, Rockford, IL, USA) in PBS at 4°C for 30 min. Cells were then rinsed three times with ice-cold 50 mM glycine in PBS (to remove unbound biotin reagent), and prepared for harvesting in general lysis buffer (1 mM sodium pyrophosphate, 50 mM Hepes, pH 7.5, 1% Triton X-100, 50 mM NaCl, 50 mM NaF, 5 mM EDTA, pH 8.0, 1 mM NaVO₄, 1 $\mu\text{g ml}^{-1}$ aprotinin,

1 $\mu\text{g ml}^{-1}$ leupeptin, 100 nM PMSF, pH 7.4). Lysates were cleared by centrifugation at 1000 g for 5 min, quantified via the Bradford method, and then a 500 μg aliquot of lysate was incubated with streptavidin cross-linked to agarose beads (Pierce) for 2 h at room temperature. Samples were then washed 4 times with general lysis buffer, and prepared for 12.5% SDS-PAGE and immunoblotting with anti-TRPV1 and/or anti-TRPA1 antibodies (Jeske *et al.* 2006).

Data analysis

For detailed statistical analysis, GraphPad Prism 4.0 (GraphPad, San Diego, CA, USA) was used. The data in figures are given as mean $\pm$ standard error of the mean (S.E.M.), with the value of *n* referring to the number of analysed cells or trials for each group. Experiments were performed at least in triplicate. Significant differences between groups were assessed by one-way analysis of variance (ANOVA) with Bonferroni's multiple comparison *post hoc* test. Two conditions were compared using paired or unpaired *t* tests. A difference was accepted as significant when $P < 0.05$, < 0.01 or < 0.001 and are identified by *, ** and ***, respectively.

Results

CAP and MO desensitize mustard oil-gated current (I_{MO}) in cultured rat TG neurons

Several compounds known to excite sensory neurons are able to activate (or inhibit) multiple TRP channels which are expressed by sensory neurons (Macpherson *et al.* 2006). Therefore, a major assumption when evaluating desensitization of receptor/channels is that the agonists exhibit receptor specificity over a defined range of concentrations. The specificity of CAP activation of TRPV1 has been shown by numerous groups (Caterina *et al.* 2000; Davis *et al.* 2000). In contrast, the selectivity of MO for TRPA1 has been disputed; a recent report claims that MO ($< 50 \mu\text{M}$) can also activate CAP-insensitive sensory neurons in TRPA1 null-mutant mice (Kwan *et al.* 2006). These responsive cells could contribute to measured desensitization of MO responses in certain experiments. Therefore, we investigated MO-induced alterations in intracellular calcium levels ([Ca²⁺]_i) in CAP-insensitive cells in cultured adult rat sensory neurons. A Ca²⁺-imaging system was employed to study large numbers of cells. From 255 recorded cells, 95 (37.2%) responded to MO (50 μM ; maximum response (E_{max}) concentration) and 122 (48%) responded to CAP (300 nM) as measured by an elevation in [Ca²⁺]_i. The vast majority (93 of 95) of the MO-sensitive cells were also responsive to CAP as measured by increased Ca²⁺ accumulation. Knock-down of TRPA1 in adult rat

sensory neurons using siRNA resulted in an approximately 75% reduction in TRPA1 expression with a corresponding 85% reduction in the number of MO responding cells (Jeske *et al.* 2006). At least in adult rat sensory neurons, MO appears to be a specific agonist for the TRPA1 channel which is expressed within the capsaicin-sensitive population of sensory neurons (Story *et al.* 2003; Jordt *et al.* 2004; Obata *et al.* 2004; Nagata *et al.* 2005).

We subsequently selected small and medium-sized (20–35 μm) TG neurons for perforated-patch recording. In this study, 109/189 (57.6%) of the tested neurons generated an inward current in response to MO (50 μM) treatment. Every MO responsive neuron ($n = 45$) also responded to CAP (300 nM). However, only 53 of 76 ($\approx 70\%$) CAP-responsive neurons responded to MO application (e.g. Figs 1A, 3A, and 4A and B). TG neurons were pre-treated once with CAP (for 30 s) or MO (for 2 min), and then at 2–3 min intervals (depending on decline of currents), a subsequent I_{MO} was recorded (Fig. 1C). It is important to note that there was a noticeable cell-to-cell variation in the size of I_{MO} and the extent of I_{MO} tachyphylaxis (Fig. 1A, C and D). As has been reported,

CAP pre-treatment elicited capsaicin gated current (I_{CAP}) tachyphylaxis in nearly all recorded neurons (Liu *et al.* 1997; Piper *et al.* 1999) (Fig. 1B and D). In contrast, the magnitude of I_{MO} tachyphylaxis was more variable, with a profound tachyphylaxis (~ 70 – 90% inhibition) observed in 6 of 13 neurons, an intermediate tachyphylaxis in 4 of 13 neurons (~ 40 – 60% inhibition) and no detectable decline in I_{MO} in 4 of 13 neurons (Fig. 1C lower panel). Interestingly, the magnitude of I_{MO} tachyphylaxis had a tendency to be dependent on the I_{MO} magnitude evoked by the first MO application. Overall, an initial pre-treatment with MO reduced the subsequent I_{MO} by 62% (Fig. 1A; -231.1 ± 31.9 pA versus -90.42 ± 9.2 pA).

The presence of 'cross' or heterologous desensitization was evaluated by pre-treating neurons with CAP followed by the measurement of I_{MO} . Since I_{MO} , like I_{CAP} , underwent tachyphylaxis, we could not identify the MO-responsive population of neurons by applying MO prior to CAP pre-treatment. Therefore, we have identified MO-responsive neurons by the presence of I_{MO} after CAP treatment. A subset of CAP-responsive neurons ($\sim 30\%$) did not reveal I_{MO} above noise. Therefore, only neurons

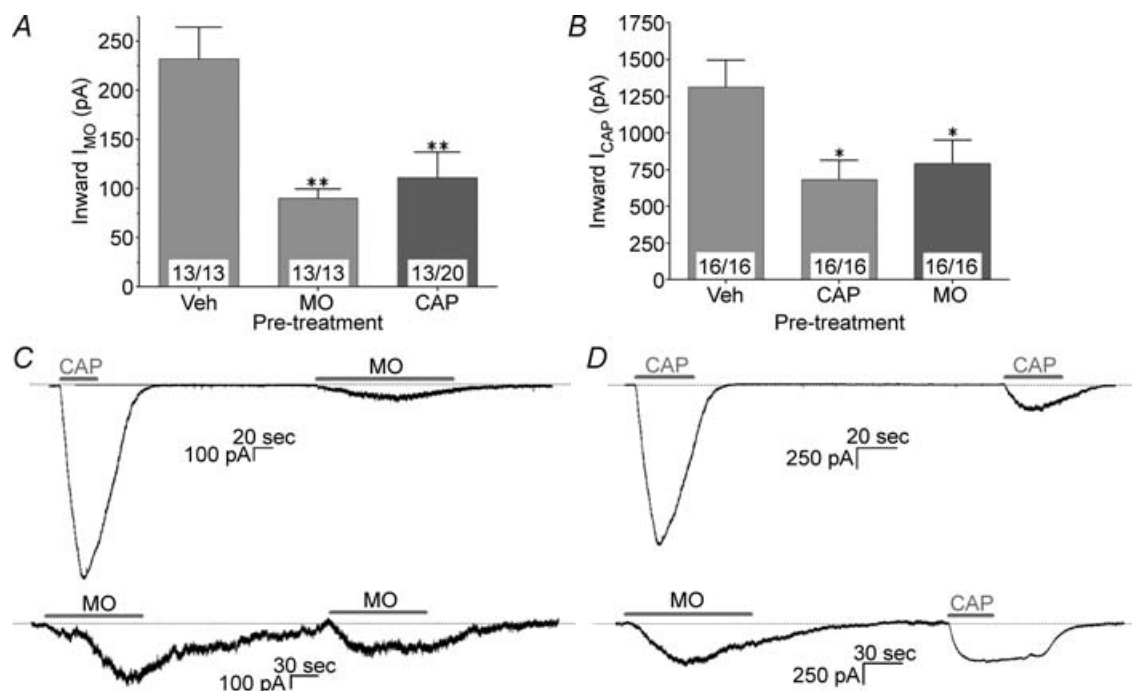


Figure 1. Cross-desensitization between TRPA1 and TRPV1 in sensory neurons

Recordings were made of trigeminal ganglia (TG) sensory neurons using perforated patch configuration. A, the desensitization of I_{MO} (50 μM ; applied for 2 min) in TG neurons pre-treated with vehicle (0.01% DMSO supplied constantly with external solution), mustard oil (MO; 50 μM for 2 min) or capsaicin (CAP; 300 nM for 30 s). B, the desensitization of I_{CAP} (300 nM applied for 30 s) in TG neurons pre-treated with vehicle, MO (50 μM) or CAP (300 nM). Drugs were applied at an interval of 3 min. The digits at the base of each bar refer to the number of cells exhibiting currents following the first challenge with drug (right part of digits; first drug is indicated on X-axis) and second application (left part of digits; second drug is indicated on Y-axis). Note: in 'vehicle group' both digits are the same number of responding cells, since there is no second drug application. Numbers of recorded/analysed cells will be represented this way in subsequent figures. C and D, representative traces are shown in the bottom panels from the I_{MO} and I_{CAP} desensitization experiments. Horizontal bars mark application of the noted drugs.

that displayed detectable I_{MO} after CAP pre-treatment were used for further analysis (Fig. 1A and C; note: 13 of 20 CAP-responsive neurons generated I_{MO} during a 2 min MO application). Figure 1A illustrates that CAP pre-treatment desensitized I_{MO} by 53% (-231.1 ± 31.9 pA versus -111.4 ± 25.5 pA). We also observed that there was no significant difference in rise time_{10–90%} of I_{MO} after MO or CAP pre-treatments (first response to MO 45.8 ± 8.0 s, $n = 13$ versus response to MO after MO or CAP pre-treatment 41.8 ± 6.7 s, $n = 26$).

We next compared the effects of MO and CAP for desensitization of I_{CAP} . Figure 1B summarizes this set of experiments. Pre-treatment with CAP desensitized I_{CAP} by 48% (first application of CAP -1311 ± 183 pA versus second application of CAP -682 ± 132 pA). Similarly, pre-treatment with MO inhibited I_{CAP} by 40% (CAP response after vehicle treatment -1311 ± 183 pA versus CAP response after MO treatment -792 ± 159 pA). In agreement with others (Liu & Simon, 1996; Piper *et al.* 1999), we found that desensitization of I_{CAP} (with MO or CAP) was accompanied by a significant change in I_{CAP} kinetics (Fig. 1D). Thus, during the first application of CAP, the rise time_{10–90%} was 8.11 ± 2.16 s ($n = 16$), while after pre-treatment with either MO or CAP, this parameter of I_{CAP} increased 3-fold (to 24.41 ± 1.68 s, $n = 32$; t test; $P < 0.001$). Acute desensitization of I_{CAP} also clearly slowed after CAP or MO pre-treatment of TG neurons (Fig. 1D). Taken together, our data suggest that, in addition to I_{MO} and I_{CAP} tachyphylaxis, MO can desensitize I_{CAP} in sensory neurons and CAP can also desensitize I_{MO} .

Pharmacological desensitization of I_{CAP} and I_{MO} in CHO cells expressing TRPA1 and/or TRPV1

We next investigated whether the desensitization of TRPA1 by MO and CAP could be observed in a heterologous expression system. TRPA1 and/or TRPV1 were expressed in CHO cells to study the heterologous desensitization of MO and CAP responses. TRPA1-containing CHO cells were not activated by CAP, nor did CAP inhibit I_{MO} (Fig. 2A and C). Moreover, TRPA1-expressing CHO cells responded only to MO application with the generation of a subsequent I_{MO} tachyphylaxis (Fig. 2A and C; see also online Supplemental Fig. 1A). Interestingly, MO treatment produced a significantly greater I_{MO} tachyphylaxis in CHO cells expressing only TRPA1 as compared to CHO cells expressing both TRPA1 and TRPV1 (Fig. 2A and online Supplemental Fig. 1A and C). Thus, MO produced a 91% reduction in I_{MO} in TRPA1-expressing CHO cells (69.31 ± 7.22 pA pF⁻¹, $n = 11$ versus 6.213 ± 1.712 pA pF⁻¹, $n = 6$), but only a 52% inhibition in TRPA1/TRPV1-expressing CHO cells (Fig. 2A). Furthermore, in 5 of 11 MO-treated CHO cells

expressing only TRPA1, virtually no currents were detected after the second application of MO.

Analogous desensitization studies were carried out in TRPV1-expressing CHO cells. In these cells, the application of MO ($50 \mu\text{M}$ for 2 min) did not generate currents above noise, while CAP (300 nM applied for 30 s) activated robust inward currents (Fig. 2D). As a result, in TRPV1-expressing CHO cells, MO pre-treatment did not desensitize I_{CAP} (Fig. 2B). In contrast, and similar to that observed in TG neurons, tachyphylaxis of I_{CAP} was observed in both TRPV1- as well as TRPA1/TRPV1-expressing CHO cells (Fig. 2D and Supplemental Fig. 1B and D; in TRPA1/TRPV1-expressing CHO cells, first CAP application 123.1 ± 14.44 pA pF⁻¹, $n = 9$ versus second CAP application 47.56 ± 12.19 pA pF⁻¹, $n = 9$).

TRPA1 and TRPV1 are greatly coexpressed in sensory ganglia (Story *et al.* 2003; Jordt *et al.* 2004; Nagata *et al.* 2005). To conduct the cell expression studies, we sought to reproduce the relative expression pattern for TRPA1 and TRPV1 observed in TG neurons (Jordt *et al.* 2004). Therefore, we cotransfected CHO cells with TRPV1 and TRPA1 (equimolar concentration; mass ratio 0.6:1, TRPV1:TRPA1). Figure 2A and B (representative traces in Fig. 2E and F) illustrate that the findings previously observed in TG neurons concerning heterologous desensitization between MO and CAP responses were essentially confirmed in these TRPV1/TRPA1-expressing CHO cells. One notable exception is the desensitization of TRPA1 by MO in TRPA1 versus TRPV1/TRPA1-expressing systems (Fig. 2A, Supplemental Fig. 1A and C). The coexpression of TRPA1 with TRPV1 substantially reduced the magnitude of I_{MO} tachyphylaxis from 91% to 52% (Fig. 2A; $P < 0.001$; two-way ANOVA), and the magnitude of this reduced effect was strikingly similar to levels observed in TG neurons (62%). Altogether, the comparison of data presented in Figs 1 and 2 demonstrates that the TRPV1/TRPA1 coexpression in CHO cells is required to reproduce the desensitization pattern observed in TG neurons.

Ca²⁺ dependency in TRPA1 desensitization

All known cellular pathways for the acute or pharmacological desensitization for TRP channels by either homologous or heterologous stimuli involve intracellular calcium accumulation (Docherty *et al.* 1996; Koplas *et al.* 1997; L. Liu *et al.* 1997; Piper & Docherty, 2000; B. Liu *et al.* 2005; Mohapatra & Nau, 2005; Nilius *et al.* 2005, 2006; Rohacs *et al.* 2005; Zhang *et al.* 2005; Zhu *et al.* 2005; Abe *et al.* 2006; Jeske *et al.* 2006). This [Ca²⁺]_i accumulation can be accomplished via direct application of Ca²⁺ to inside-out patches (Nilius *et al.* 2005, 2006; Zurborg *et al.* 2007) or via extracellular Ca²⁺

entry mediated by calcium-permeable TRP channels (Docherty *et al.* 1996; Nagata *et al.* 2005; Nilius *et al.* 2005; Rohacs *et al.* 2005; Zhu *et al.* 2005). Since TRPA1 and TRPV1 are calcium-permeable channels (Docherty *et al.* 1996; Caterina *et al.* 2000; Bandell *et al.* 2004; Jordt *et al.* 2004), there is a reasonable expectation that the pharmacological desensitization of TRPA1 or TRPV1 by MO and CAP might be triggered by elevations in $[Ca^{2+}]_i$. In agreement with previous findings (Docherty *et al.* 1996; Koplas *et al.* 1997), the removal of extracellular Ca^{2+} abolished the development of an I_{CAP} tachyphylaxis (see online Supplemental Fig. 2A and B). In addition, in a Ca^{2+} -free solution, the I_{CAP} decay time, but not activation time, was significantly slower (Fig. 1C and D, *versus* Supplemental Fig. 2B and C). This finding is also in agreement with previous studies that have demonstrated Ca^{2+} dependence for the acute desensitization of I_{CAP} (Docherty *et al.* 1996; Koplas *et al.*

1997). The pharmacological desensitization of TRPV1 by MO also required extracellular Ca^{2+} (Supplemental Fig. 2A and C). The use of a Ca^{2+} -free solution had the following effects on I_{MO} properties in sensory neurons: (1) The magnitude of I_{MO} was significantly larger in Ca^{2+} -free solution *versus* that observed in SES (Figs 1 and 3; in SES -232.1 ± 31.92 pA, $n = 13$ *versus* in Ca^{2+} -free -457.2 ± 98.93 pA, $n = 18$; $P < 0.01$ *t* test). In contrast, in TRPA1 heterologous systems, I_{MO} is larger in the presence of extracellular Ca^{2+} (Jordt *et al.* 2004; Nagata *et al.* 2005), and this might be related to recent observations that intracellular Ca^{2+} ($> 1 \mu M$) activates TRPA1 (Doerner *et al.* 2007; Zurborg *et al.* 2007). (2) The deactivation decay time of I_{MO} after removal of the agonist in Ca^{2+} -free solution was significantly prolonged (traces in Figs 1 and 3, and Supplemental Fig. 2; in SES 63.8 ± 12.0 s, $n = 29$ *versus* in Ca^{2+} -free solution 240.1 ± 42.3 s, $n = 28$; $P < 0.05$ *t* test). This effect was also observed in

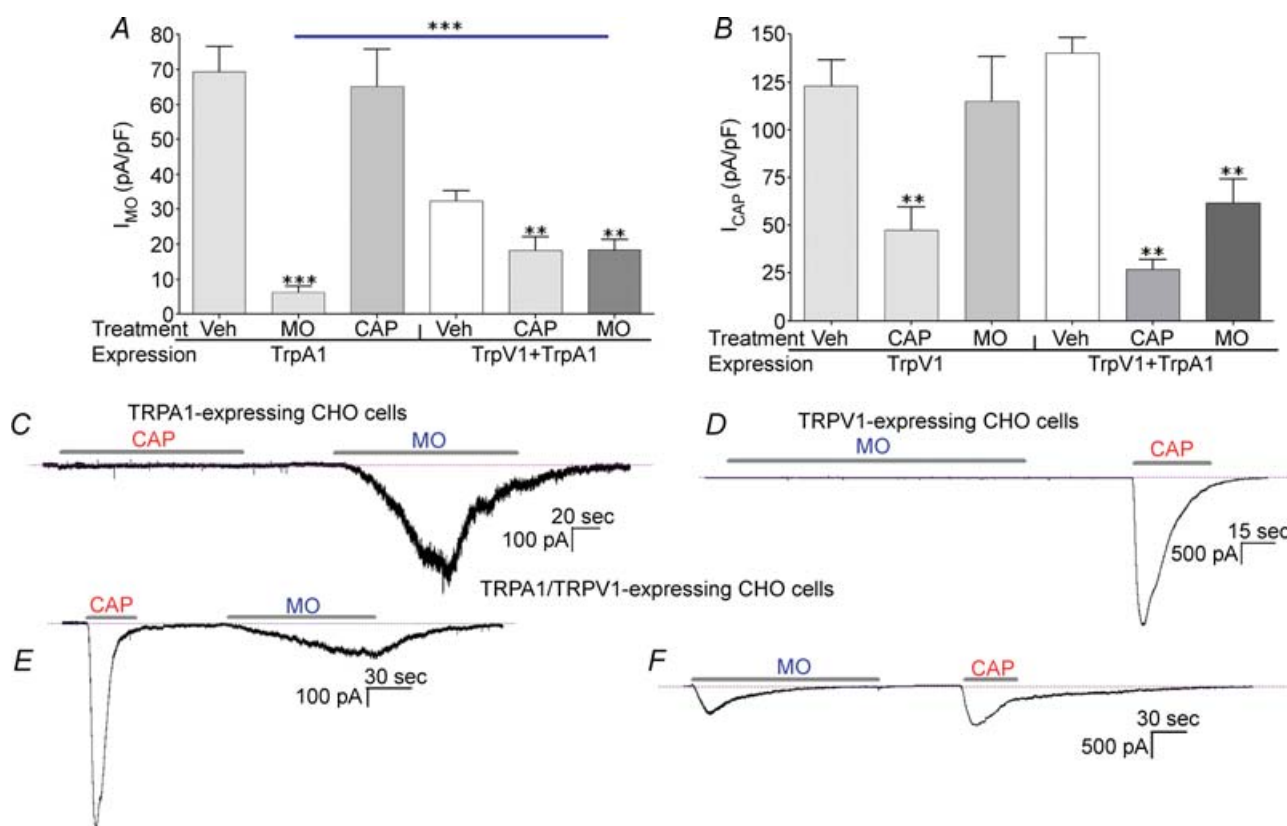


Figure 2. Cross-desensitization between TRPA1 and TRPV1 in CHO cells

Recordings were made of TRPA1 and/or TRPV1-expressing CHO cells using perforated patch configuration. A, pharmacological desensitization of I_{MO} ($50 \mu M$) induced by pre-treatment with CAP ($0.3 \mu M$) or MO ($50 \mu M$) was determined in transiently transfected CHO cells. B, pharmacological desensitization of I_{CAP} (300 nM) evoked by application of CAP and MO in CHO cells. The Y-axis reflects current density ($pA pF^{-1}$) for I_{MO} (A) and I_{CAP} (B), while the X-axis indicates drugs for pre-treatment and the types of channels expressed in the CHO cells. Drugs were applied at an interval of 3 min; CAP was applied for 30 s and MO was applied for 2 min. C and D, representative traces of I_{MO} and I_{CAP} recorded from TRPV1- (C) and TRPA1- (D) expressing CHO cells. E and F, representative traces of CAP-induced I_{MO} (E) and MO-induced I_{CAP} (F) desensitization in TRPA1/TRPV1-coexpressing CHO cells. Horizontal bars mark application of the drugs.

TRPA1-expressing cells (Macpherson *et al.* 2007). This phenomenon could be explained by a slow reversal of covalent modification of reactive cysteines within TRPA1 by MO in absence of extracellular Ca^{2+} (Macpherson *et al.* 2007). (3) Importantly, it appears that a Ca^{2+} -free extracellular solution had little, if any, detectable effect on the inactivation (i.e. acute desensitization) of I_{MO} within 2 min of MO application (traces in Fig. 3 and Supplemental Fig. 2). The acute desensitization time of I_{MO} in sensory neurons proved to be difficult to quantify, because in the majority of neurons (> 65%), I_{MO} inactivation was less than 20% during a 2 min MO application in either the presence or absence of extracellular Ca^{2+} . Interestingly, and in agreement with previous reports (Jordt *et al.* 2004; Nagata *et al.* 2005), the inactivation of I_{MO} , at least in the presence of extracellular Ca^{2+} , was detected in TRPA1- as well as TRPA1/TRPV1-expressing CHO cells (Fig. 2C and F, Supplemental Fig. 1A and C). Altogether these results

indicate that (1) the pharmacological desensitization of TRPV1 with MO, like CAP, was Ca^{2+} dependent in sensory neurons. (2) The peak value of I_{MO} recorded from sensory neurons is greater in a Ca^{2+} -free solution. (3) However, the I_{MO} inactivation kinetics in sensory neurons, unlike TRPA1-expressing CHO cells, does not appear to be noticeably affected by the use of Ca^{2+} -free solution.

We next examined the Ca^{2+} dependency of pharmacological desensitization of TRPA1 by pre-treatment with CAP or MO. In the absence of Ca^{2+} , CAP was not able to desensitize TRPA1 (Fig. 3A and E). In contrast, MO continued to significantly desensitize TRPA1 in Ca^{2+} -free conditions, although the magnitude of MO-induced TRPA1 pharmacological desensitization was influenced by the removal of external Ca^{2+} . Thus, MO pre-treatment desensitized I_{MO} by 62% in the presence of calcium and by 47% in the absence of calcium (Figs 1A and 3A). Statistically, such a difference

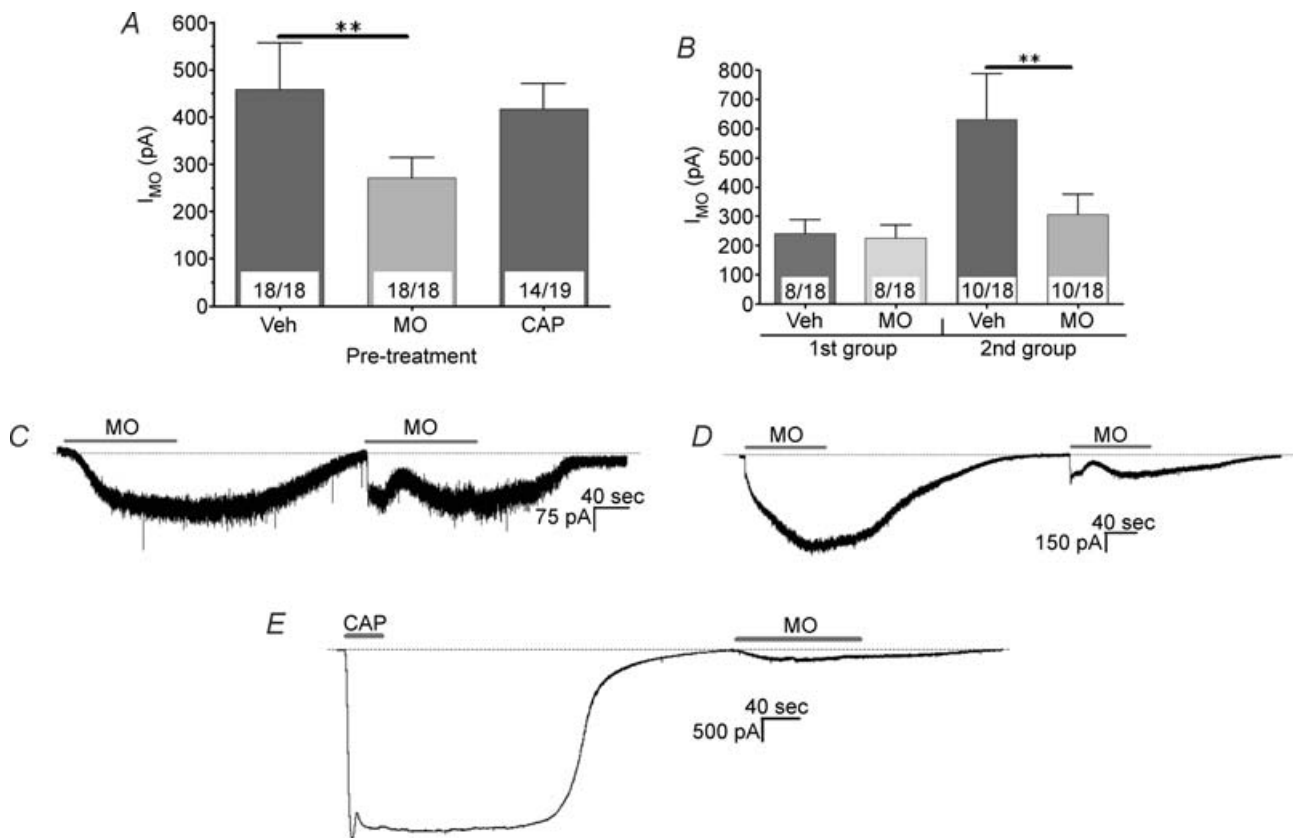


Figure 3. Effect of extracellular Ca^{2+} on the desensitization of TRPA1 by MO and CAP in sensory neurons

Recordings were made of sensory neurons using perforated patch configuration. A, summary data for I_{MO} desensitization by CAP and MO in Ca^{2+} -free extracellular solution. CAP (300 nM; for 30 s) or MO (50 μM ; for 2 min) was applied at an interval of 3–8 min. I_{MO} and I_{CAP} have slow decay times after drug removal. Therefore, if currents did not decline to baseline within 3 min of beginning the treatment, then the second application was delayed for 2–5 min (see C and D). B, separate presentation of I_{MO} tachyphylaxis in Ca^{2+} -free extracellular solution for two types of neurons. In the first group ($n = 8$), I_{MO} tachyphylaxis is not registered, while in the second group ($n = 10$), tachyphylaxis is observed. C and D, representative traces illustrate I_{MO} tachyphylaxis for neurons belonging to the first (C) and second (D) groups. E, trace shows lack of I_{MO} desensitization by CAP in Ca^{2+} -free conditions. Horizontal bars mark application of the drugs.

is insignificant; however, two types of MO-sensitive neurons could be described from these experiments. In the first group ($n = 8$ of 18) of neurons, the magnitude of the initial I_{MO} was smaller compared to the second group (1st group, -241.1 ± 46.61 pA, $n = 8$ versus 2nd group, -630.2 ± 156.4 pA, $n = 10$; $P < 0.01$ t test) and the subsequent I_{MO} did not undergo tachyphylaxis in Ca^{2+} -free media (Fig. 3B and C). In contrast, in the second group ($n = 10$ of 18), I_{MO} was desensitized in Ca^{2+} -free media (Fig. 3B and D). As described above, approximately 30% (4 of 13) of neurons were resistant to I_{MO} tachyphylaxis even in the presence of external Ca^{2+} . Therefore, we can estimate that in $\sim 85\%$ (i.e. $30\% + 55\%$) of neurons I_{MO} desensitization is Ca^{2+} -independent (Figs 1A and 3B). Note that this value is only estimation, since experiments were performed on separate pools of neurons. To summarize this set of experiments, TRPA1 in TG sensory neurons can be desensitized through two pathways, consisting of either Ca^{2+} -dependent (by CAP) or Ca^{2+} -independent (by MO) mechanisms.

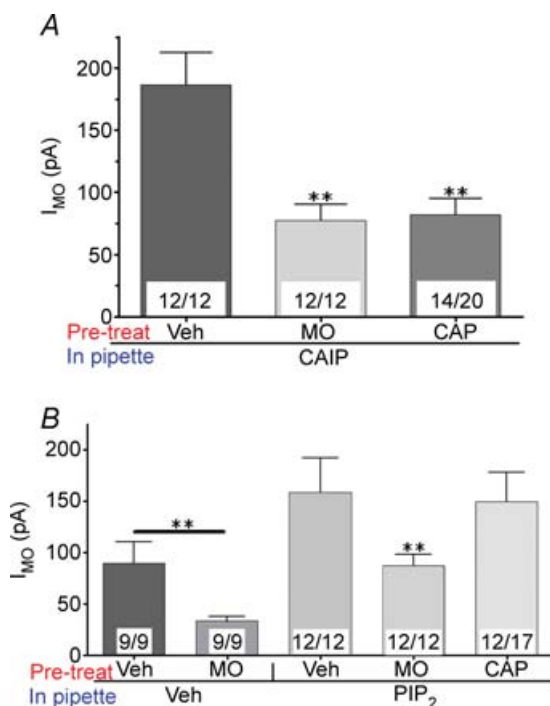


Figure 4. Roles of calcineurin and PIP₂ biosynthesis pathways in pharmacological desensitization of TRPA1 by CAP and MO in sensory neurons

Recordings were made of sensory neurons using whole-cell patch voltage clamp ($V_h = -60$ mV) configuration without ATP and GTP in the pipette solution. A, treatment of sensory neurons with the calcineurin inhibitor, CAIP, did not prevent TRPA1 desensitization by CAP and MO. CAIP ($50 \mu M$) was dialysed into cells via the recording pipette for 5 min. B, dialysis of sensory neurons for 15 min with natural PIP₂ ($100 \mu M$) prevented desensitization of TRPA1 by CAP, but not by MO.

Calcineurin does not play a role in pharmacological desensitization of the TRPA1 channel

Calcineurin is activated by an increased accumulation of $[Ca^{2+}]_i$ and there is evidence pointing to an important role for calcineurin in the acute as well as pharmacological desensitization of TRPV1 (Docherty *et al.* 1996; Mohapatra & Nau, 2005; Patwardhan *et al.* 2006). We next investigated whether the pharmacological desensitization of TRPA1 is mediated by the calcineurin pathway. Our initial experiments evaluated I_{CAP} tachyphylaxis as a positive control to test the efficacy of the calcineurin antagonist, calcineurin autoinhibitory peptide (CAIP). CAIP ($50 \mu M$; Bachem) was included in the pipette solution and delivered into the cells by 5 min dialysis in whole-cell patch configuration. As reported (Piper *et al.* 1999), the magnitude of I_{CAP} was reduced by the presence of CAIP in the pipette. The I_{CAP} at first delivery of CAP application was approximately 60% of I_{CAP} recorded under perforated patch configurations (Fig. 1B and Supplemental Fig. 3A). In agreement with results of previous studies (Piper *et al.* 1999), pre-treatment of cells with CAIP effectively prevented I_{CAP} tachyphylaxis (Supplemental Fig. 3A). We next performed similar types of experiments to study the pharmacological desensitization of TRPV1 by MO. Supplemental Fig. 3 shows that MO-triggered inhibition of I_{CAP} could also be reversed by the calcineurin antagonist. Since I_{MO} tachyphylaxis is independent of external Ca^{2+} , it was expected that TRPA1 desensitization by MO would be insensitive to CAIP. Indeed, treatment of TG neurons with CAIP had no significant effect on I_{MO} tachyphylaxis (Fig. 4A). Furthermore, blockage of calcineurin activity did not affect the magnitude of I_{MO} (Fig. 1A and Fig. 4A). The results demonstrated in Fig. 4A also show that the calcineurin antagonist was ineffective in preventing CAP-induced pharmacological desensitization of TRPA1. Taken together, the results demonstrate that TRPV1 desensitization by either CAP or MO is mediated through the calcineurin pathway. These findings are in good agreement with earlier reports that acute or pharmacological desensitization of TRPV1, mediated by either homologous or heterologous mechanisms is achieved via the calcineurin pathway (Docherty *et al.* 1996; Piper *et al.* 1999; Mohapatra & Nau, 2005; Patwardhan *et al.* 2006). In contrast, the inhibition of TRPA1 activities by either CAP or MO treatment of neurons was not affected by calcineurin suppression.

PIP₂ depletion regulates Ca^{2+} -dependent desensitization of TRPA1

PIP₂ is a general regulator of numerous TRP channels (Clapham, 2003). Previous studies have demonstrated that the acute or pharmacological desensitization of the

TRPM4 (Nilius *et al.* 2006), TRPM5 (Nilius *et al.* 2005), TRPM8 (Rohacs *et al.* 2005), TRPV1 (Liu *et al.* 2005), TRPV5 (Rohacs *et al.* 2005) and TRPV6 (Nilius *et al.* 2005) channels is controlled by Ca^{2+} -dependent PIP_2 depletion and/or replenishment in heterologous expression systems. Accordingly, we investigated whether PIP_2 is involved in the regulation of TRPA1 pharmacological desensitization by CAP or MO in TG neurons. Initially, we attempted to block PLC activity with the widely used inhibitor U73122 ($10\text{ }\mu\text{M}$). However, as previously reported (Bonnington & McNaughton, 2003), U73122 displayed unacceptable side-effects in sensory neurons: specifically, the generation of a long-lasting 'leak' current and/or an increase in resting $[\text{Ca}^{2+}]_i$ levels. In an alternative approach presented here, natural $\text{PI}(4,5)\text{P}_2$ isolated from bovine brain (Avanti Polar Lipid, Alabaster, AL, USA) was employed since it is the major physiological phosphoinositide regulator (Fruman *et al.* 1998) and is more abundant in the plasma membrane than $\text{PI}(3,4)\text{P}_2$ and $\text{PI}(3,4,5)\text{P}_3$. Natural PIP_2 can be solubilized in aqueous solutions by sonication for 5–10 min, yet it remains membrane impermeable (Gamper *et al.* 2004; Liu & Qin, 2005). Therefore, natural PIP_2 ($100\text{ }\mu\text{M}$) was applied to the inner surface of neurons by dialysis through the recording pipette (Gamper *et al.* 2004; Liu *et al.* 2005).

To evaluate the potential role of PIP_2 in the pharmacological desensitization of TRPA1, we first employed a positive control to verify the activity of natural PIP_2 dialysed into sensory neurons. Previous studies have shown that $\text{G}_{q/11}$ -protein activation by bradykinin (BK) or the protease receptor (PAR-2) agonist SLIGR leads to degradation of membrane-bound PIP_2 , and in turn, inhibition of N-type and P/Q-type Ca^{2+} channels (Wu *et al.* 2002; Gamper *et al.* 2004), which are major components of high voltage-activated (HVA) Ca^{2+} currents (I_{Ca}) in sensory neurons (Nowycky *et al.* 1985; Grigaliunas *et al.* 2002). Therefore, as a positive control, we dialysed PIP_2 into sensory neurons and evaluated the blockade of BK (or SLIGR)-induced inhibition of HVA I_{Ca} . BK and SLIGR effectively inhibited I_{Ca} in approximately 30% (7/19) of sensory neurons (Supplemental Fig. 4A and C). The inhibition of I_{Ca} typically did not recover within 8–10 min (Supplemental Fig. 4B). However, the inclusion of natural PIP_2 ($100\text{ }\mu\text{M}$) in the pipette solution prevented BK or SLIGR inhibition of HVA I_{Ca} (Supplemental Fig. 4A, B and D). It is worth noting that, in the presence of PIP_2 , some neurons ($n = 2$) still exhibited modest ($\sim 20\%$) inhibition of I_{Ca} by BK. In summary, natural PIP_2 was delivered as effectively as its synthetic analogue, diC8- PIP_2 (Wu *et al.* 2002; Gamper *et al.* 2004), into cells leading to a blockade of BK-evoked inhibition of HVA I_{Ca} in sensory neurons.

In the present study, we characterized the desensitization of I_{MO} recorded in perforated patch configuration. Since PIP_2 delivery required conducting

recordings in whole-cell configuration after prolonged dialysis (15 min), we first evaluated I_{MO} tachyphylaxis after 15 min of establishing whole-cell configuration. Figure 4B illustrates that in whole-cell configuration, I_{MO} underwent the same pronounced desensitization as observed previously with perforated patch (64% *versus* 62%). However, the magnitude of the initial I_{MO} was as much as 2.5-fold larger than when recording was performed in perforated patch configuration (perforated patch, $-231.1 \pm 31.9\text{ pA}$, $n = 13$ *versus* whole-cell, $-89.76 \pm 20.67\text{ pA}$, $n = 9$; $P < 0.01$). The dialysis of sensory neurons with PIP_2 ($100\text{ }\mu\text{M}$) for 15 min resulted in partial restoration of the amplitude of the initial I_{MO} ($-158.3 \pm 33.51\text{ pA}$, $n = 12$). Importantly, the delivery of PIP_2 from the recording pipette prevented CAP-evoked pharmacological desensitization of TRPA1 (Fig. 4B). In contrast, treatment of sensory neurons with PIP_2 did not significantly alter I_{MO} tachyphylaxis (Fig. 4B).

A recent report has claimed that TRPV1 tachyphylaxis in a heterologous expression system is regulated by PIP_2 (Liu *et al.* 2005). We therefore investigated the potential role of PIP_2 in the pharmacological desensitization of TRPV1 by MO or CAP in sensory neurons. Control recordings in the whole-cell configuration demonstrated that I_{CAP} undergoes tachyphylaxis with a higher magnitude compared to I_{CAP} recording performed in perforated patch configuration (78% *versus* 40%; Supplemental Fig. 3B). The 15 min dialysis of PIP_2 into sensory neurons slightly attenuated the amount of pharmacological desensitization of TRPV1 by CAP or MO; however, TRPV1 inhibition persisted under these conditions (Supplemental Fig. 3B). Altogether, these results indicate that: (1) CAP-, but not MO-, induced desensitization of TRPA1 is mediated by hydrolysis of PIP_2 ; and (2) the pharmacological desensitization of TRPV1 in sensory neurons does not involve PIP_2 depletion.

CAP- and MO-evoked PIP_2 depletion in CHO cells and sensory neurons

Ca^{2+} influx through TRP channels could activate Ca^{2+} -sensitive PLC isoforms (Rohacs *et al.* 2005). We next investigated whether Ca^{2+} influx through TRPA1 and TRPV1 activates a Ca^{2+} -sensitive PLC that depletes PIP_2 , thus causing CAP-induced desensitization of TRPA1 channels. To examine PLC activation, we measured the translocation of the GFP-tagged PLC_δ pleckstrin homology domain (PLC_δ -PHD) as a biosensor for enzyme activity. The expression of PLC_δ -PHD in cultured cells leads to its binding of PIP_2 and corresponding localization of the construct product to the membrane. Since the affinity of PLC_δ -PHD for IP_3 is ~ 10 -fold higher than for PIP_2 , the product of the construct serves as a real-time biosensor for PLC hydrolysis of membrane-bound PIP_2 .

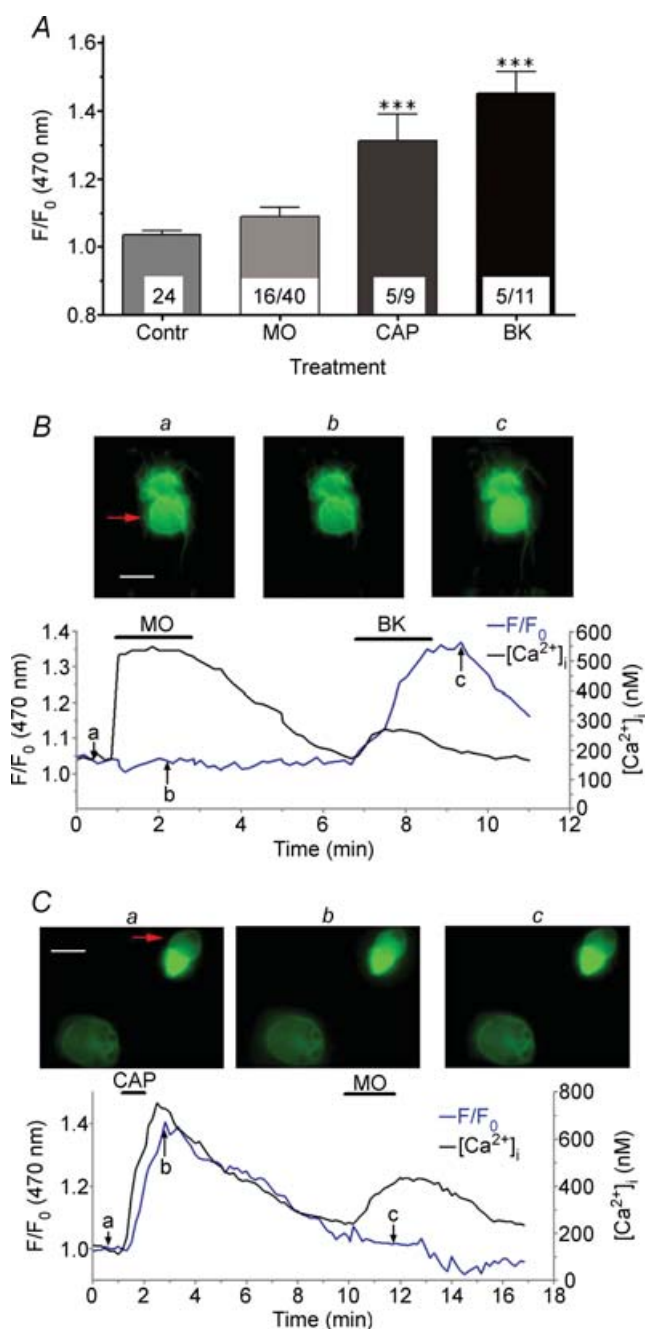


Figure 5. CAP, but not MO, activates phospholipase C (PLC) and hydrolyses PIP₂ in sensory neurons

A, CAP (300 nM) and bradykinin (BK; 100 nM), but not MO (50 μ M), treatments of sensory neurons expressing GFP-tagged PLC δ -PHD caused translocation of the GFP-PLC δ -PHD from the plasma membrane region into the cytosol. Fluorescence emission (at 470 nm) was measured in the cytoplasmic area of neurons. The Y-axis represents normalized data (F/F_0) against baseline emission density of the same area (F_0). The digits inside the bars indicate the number of responsive neurons (left side) and the total number of analysed neurons (right side). The negative control in this experiment was the measurement of GFP-tagged PLC δ -PHD translocation in MO non-responsive cells (defined by MO application producing no accumulation in $[Ca^{2+}]_i$). B and C, representative simultaneous real-time fluorescence imaging traces of $[Ca^{2+}]_i$ accumulation (black

into soluble IP₃ by measuring cytosolic translocation as indicated by changes in fluorescence in the peri-membrane region. It is probable that PLC δ -PHD translocation reflects both PIP₂ depletion and IP₃ production (Hirose *et al.* 1999), and the relative contributions of PIP₂ and IP₃ in the translocation of PLC δ -PHD have been debated (van der Wal *et al.* 2001); however, it is generally agreed that translocation of PLC δ -PHD into the cytosol reflects PLC activation and PIP₂ hydrolysis (Hirose *et al.* 1999; Gamper *et al.* 2004; Rohacs *et al.* 2005).

In these experiments, MO or CAP was applied to sensory neurons containing the PLC δ -PHD; and then both $[Ca^{2+}]_i$ accumulation and GFP-tagged PLC δ translocation was recorded in alternating measurements (Fig. 5B and C). The lack of translocation in MO-insensitive and PLC δ -PHD-containing cells served as a negative control, while BK-evoked redistribution of GFP-tagged PLC δ was a positive control. Figure 5A summarizes the results of these experiments. In MO-insensitive cells, no translocation was registered following MO treatment (i.e. F/F_0 value did not increase above 1.15). In 13 of 16 MO-responsive neurons, translocation was not observed, despite an elevation in $[Ca^{2+}]_i$ and the detection of subsequent BK-evoked PLC δ translocation (Fig. 5A and B). Interestingly, our previous experiment demonstrated 3 of 18 neurons displaying I_{MO} tachyphylaxis that was Ca^{2+} dependent (Fig. 3); this ratio is similar to the percentage of neurons (3 of 16; $\approx 20\%$) in which MO-evoked PLC δ -PHD translocation was observed. Unlike pre-treatment with MO, CAP treatment of neurons ignited a robust translocation of PLC δ -PHD in every CAP-positive neuron (Fig. 5A and C). This is indicative of PIP₂ hydrolysis, presumably resulting from activation of a Ca^{2+} -sensitive PLC triggered by TRPV1-mediated Ca^{2+} influx into neurons. The CAP-evoked Ca^{2+} influx was generally larger than that induced by MO (for CAP: 1057 ± 164.3 nM, $n = 5$ versus for MO: 666.1 ± 63.75 nM, $n = 16$; t test $P < 0.01$;

line) and GFP-PLC δ -PHD translocation (blue line) after treatment of neurons with MO followed by BK (B) or CAP followed by MO (C). Data were collected every 10 s with alternating exposure for 200 ms at 340 nm, for 200 ms at 380 nm and for 300 ms at 470 nm. Horizontal bars mark the duration of indicated drug applications. Fluorescence images highlight distribution of GFP-PLC δ -PHD in neurons at time points a, b or c. GFP-PLC δ -PHD is mainly distributed on the plasma membrane at the time points a and b, while at the time point c, GFP-PLC δ -PHD translocates from the plasma membrane to the cytosol, leading to an increase in fluorescence density in the cytosolic area (measurement area) of the cells and a reduction in fluorescence on the rim of the cell. Note that this effect is detected in the lower, but not the upper neuron. Baseline readings were collected for 1 min, after which CAP or MO was applied for 40 s or 2 min, respectively. The neurons were washed for 4 min and BK (as a positive control) or MO (to demonstrate CAP-evoked desensitization of TRPA1) was reapplied for 2 min each. Both experimental protocols demonstrated MO-unresponsive neurons. Responsive cells are marked by red arrows. White scale bars represent 20 μ m.

Fig. 5C). The greater amount of Ca^{2+} influx generated by CAP may be related to an increase in PLC_δ -PHD translocation, although alternative possibilities could be considered (see Discussion).

Studies on PIP_2 -based regulation of TRP channel desensitization have been primarily carried out in heterologous expression systems (Liu *et al.* 2005; Nilius *et al.* 2005, 2006; Rohacs *et al.* 2005). Over-expressing channels could generate a larger and faster influx of Ca^{2+} than that which is observed in sensory neurons. In addition, the cellular biochemistry of cell lines, such as COS, HEK and CHO, could differ from sensory neurons in terms of key signalling proteins, and it is possible that sensory neurons cultured under particular conditions could have fewer Ca^{2+} -activated PLC isoforms. Therefore, we evaluated PLC activation and PIP_2 hydrolysis by CAP and MO in CHO cells expressing TRPA1 or TRPA1/TRPV1. Data reflected in Fig. 6A and C demonstrate that CAP can effectively hydrolyse PIP_2 in TRPV1/TRPA1-expressing CHO cells and in sensory neurons (Fig. 5A). However, major differences in PIP_2 depletion were noted after MO treatment in these expression systems. Unlike the case in cultured sensory neurons (Fig. 5A), MO evoked a significant PLC activation in both TRPA1- and TRPA1/TRPV1-containing CHO cells (Fig. 6A and B). Indeed, both CAP (300 nM) and MO (50 μM) generated a substantially higher rise in $[\text{Ca}^{2+}]_i$ in CHO cells than in sensory neurons (for MO: 1373 ± 285.6 nM, $n = 18$, t test $P < 0.001$ in TRPA1-expressing CHO cells or 993 ± 153.3 nM, $n = 11$, t test $P < 0.01$ in TRPA1/TRPV1-expressing CHO cells *versus* 666.1 ± 63.75 nM, $n = 16$ in TG neurons). Since the MO-induced elevation in $[\text{Ca}^{2+}]_i$ in CHO cells was comparable to the CAP-induced increase in $[\text{Ca}^{2+}]_i$ in sensory neurons, PIP_2 depletion by MO in TRPA1-expressing CHO cells could be attributed to the generation of a sufficient (i.e. suprathreshold) Ca^{2+} influx (see Discussion). Alternatively, Ca^{2+} -activated PLC isoforms could be more abundant in CHO cells than in sensory neurons. Taken together, our results indicate that MO is capable of activating PLC and depleting PIP_2 in TRPA1-overexpressing CHO cells, but not in sensory neurons. Meanwhile, CAP can effectively hydrolyse PIP_2 in TRPV1/TRPA1-expressing CHO cells and in sensory neurons.

TRPV1 regulates the pharmacological desensitization of TRPA1 by MO

Our results indicate that the magnitude of MO-induced pharmacological desensitization of TRPA1 is significantly greater in TRPA1-expressing CHO cells compared to TRPA1/TRPV1-expressing CHO cells (Fig. 2A). TRPA1 is also extensively coexpressed with TRPV1 in sensory

neurons (Kobayashi *et al.* 2005; Obata *et al.* 2005). Consistent with this observation, our data demonstrate that MO responses in adult rat TG neurons were almost always recorded from CAP-sensitive cells. To examine whether the rate of TRPA1 desensitization was altered by the presence or absence of TRPV1, we characterized I_{MO} tachyphylaxis in sensory neurons cultured from wild-type (WT) and TRPV1 null-mutant (KO) mice. The magnitude of I_{MO} was slightly larger in sensory neurons from wild-type compared to those from TRPV1 null-mutant mice (Fig. 7A). As in the construct-expressing CHO cells (Fig. 2A), I_{MO} underwent a significantly greater magnitude of tachyphylaxis in sensory neurons lacking TRPV1 expression (Fig. 7A and C; in WT, 53% *versus* in KO, 83%, two-way ANOVA $P < 0.001$). In WT mice, I_{MO} tachyphylaxis was detected in most, but not all neurons; whereas, in the TRPV1 KO mice, TRPA1 desensitization by MO was observed in every characterized cell. Furthermore, in 3 of 11 neurons isolated from TRPV1 KO TG, no I_{MO} was detected after a second application of MO. Taken together, these data suggest that MO-evoked pharmacological desensitization of TRPA1 in sensory neurons is significantly reduced by the coexpression of TRPV1.

To further explore the possibility that I_{MO} tachyphylaxis is Ca^{2+} independent, we studied extracellular Ca^{2+} involvement in TRPA1 desensitization in sensory neurons from WT and TRPV1 KO mice. Analysis of these recordings showed that the peak values of I_{MO} in Ca^{2+} -free external solution were significantly larger in neurons from WT mice (SES: -178.7 ± 27.14 pA *versus* Ca^{2+} -free: -370 ± 56.45 pA, two-way ANOVA $P < 0.05$). I_{MO} in TRPA1-overexpressing systems is larger in the presence of Ca^{2+} because TRPA1 could be activated by large ($> 1 \mu\text{M}$) intracellular Ca^{2+} accumulation triggered by MO (Jordt *et al.* 2004; Nagata *et al.* 2005; Doerner *et al.* 2007; Zurborg *et al.* 2007). However, in sensory neurons, the removal of external Ca^{2+} did not significantly affect the magnitude of I_{MO} in TRPV1 KO mice (SES: -121.5 ± 20.64 pA *versus* Ca^{2+} -free: -135.5 ± 34.95 pA, two-way ANOVA, non-significant (n.s.)). These data are consistent with the hypothesis that TRPV1 regulates the pharmacological desensitization of TRPA1. The results summarized in Fig. 7A also demonstrate that I_{MO} tachyphylaxis still persists in both WT and KO mouse neurons after the removal of external Ca^{2+} . As observed in rat TG neurons, two types of neurons could be outlined (Fig. 7B): the first group of neurons exhibited no I_{MO} tachyphylaxis, while a pronounced I_{MO} desensitization was observed in the second group of neurons (Fig. 7D). In Ca^{2+} -free conditions, the number of neurons in which I_{MO} tachyphylaxis did not occur was similar in both mouse lines (Fig. 7B; in WT: 5 of 12 *versus* in KO: 4 of 11). However, the magnitude of I_{MO} tachyphylaxis in the second group of neurons was more pronounced in KO

mice (in WT, 37% versus in KO, 63%), because the overall TRPA1 desensitization rate is greater in TG neurons isolated from these animals (Fig. 7A). Collectively, these data suggest that the presence of TRPV1 increases the magnitude of pharmacological desensitization of TRPA1 in sensory neurons.

In general, the pharmacological desensitization of channels can occur either by long-lasting changes in biophysical properties or by a stimulus-dependent fast

internalization of channels, which could be triggered via a variety of signalling cascades including those which are Ca^{2+} dependent (Bueno *et al.* 1998; Wu *et al.* 2006). Further, internalization of membrane-bound channels can be suppressed or even prevented by selective binding interactions with other proteins, including subunits of the channels (Jugloff *et al.* 2000; Bernstein & Jones, 2006). Therefore, one possible hypothesis that predicts the observation that the desensitization of TRPA1 is

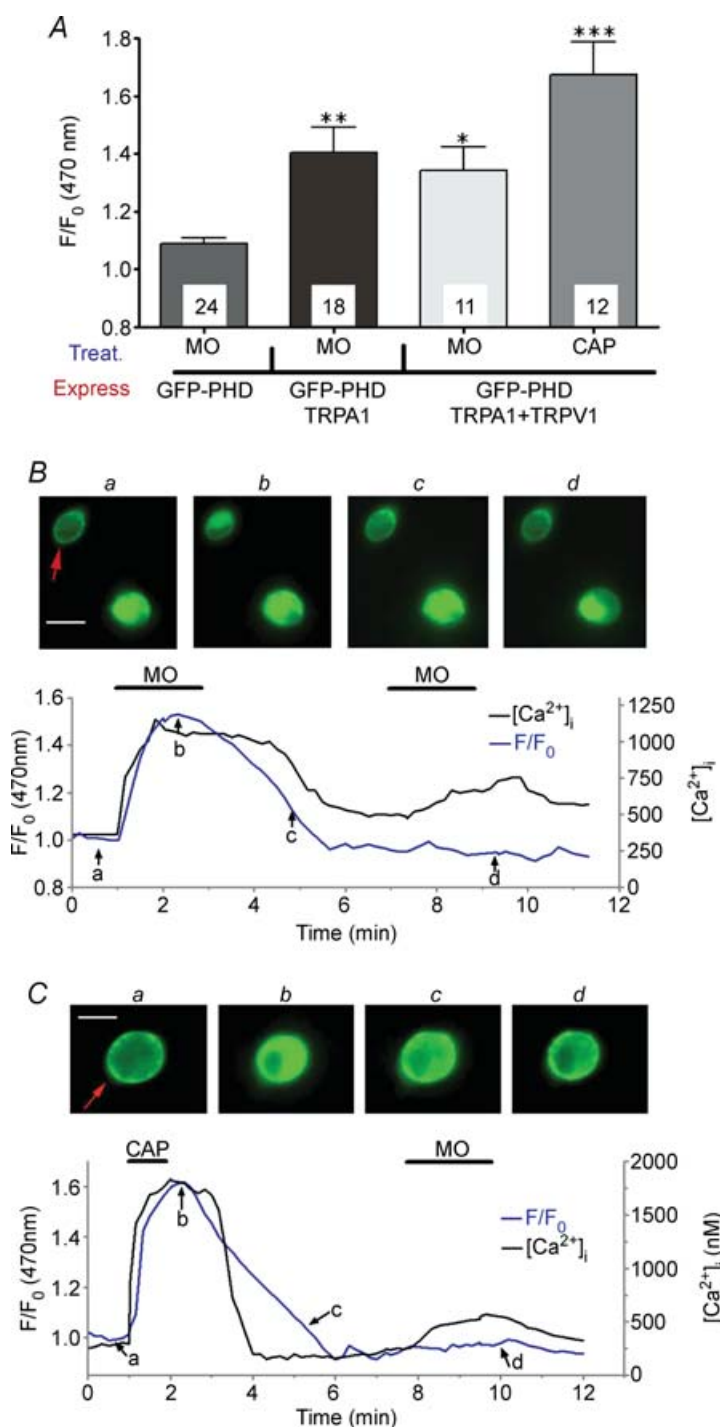


Figure 6. CAP hydrolyses PIP_2 in TRPA1/TRPV1-expressing CHO cells, and MO hydrolyses PIP_2 in both TRPA1 and TRPA1/TRPV1-expressing CHO cells

A, effect of CAP (300 nM) and MO (50 μM) treatments of CHO cells expressing either TRPA1/GFP-PLC δ -PHD or TRPA1/TRPV1/GFP-PLC δ -PHD on the translocation of PLC δ -PHD. CHO cells expressing GFP-PLC δ -PHD alone were considered as negative controls. The types of treatment and expression patterns of CHO cells are noted below the X-axis. B and C, representative fluorescence imaging traces of TRPA1/TRPV1/GFP-PLC δ -PHD-expressing CHO cells treated twice with MO (B) or with CAP followed by MO (C). Experiments were performed as described in the legend for Fig. 5. Responsive cells are marked by red arrows. White scale bars represent 7 μm .

reduced in the presence of TRPV1 is based on TRPV1 stabilizing the expression of the TRPA1 channel at the cell surface. To test this notion, we evaluated the effect of the TRPV1 channel on MO-induced TRPA1 internalization in TRPA1- and TRPA1/TRPV1-expressing CHO cells as well as in sensory neurons. After MO treatment, biotinylated cell surface proteins were purified with streptavidin–agarose, and probed for TRPA1 and TRPV1 immunoreactivity with specific anti-TRPA1 and anti-TRPV1 antibodies. The specificities of both antibodies were thoroughly demonstrated in previous studies (Jeske *et al.* 2006). Aliquots (50 μ l) taken from cell lysates before pull-down with streptavidin were used as normalization controls (Figs 8 and 9). The application of MO produced a significant $27.0 \pm 4.7\%$ ($n = 3$) reduction

in cell surface levels of TRPA1 in CHO cells expressing only TRPA1 (Fig. 8A), and this effect was not observed in CHO cells expressing both TRPA1 and TRPV1 (Fig. 8D). In contrast, MO had no effect on cell surface levels of TRPV1 in CHO cells expressing either TRPV1 alone (Fig. 8B) or expressing both TRPA1 and TRPV1 (Fig. 8E). To demonstrate the specificity of labelling only surface proteins with biotin in transfected CHO cells and sensory neurons, we probed for an intracellular protein, β -actin, as a negative control (Figs 8C and 9C).

We next examined whether MO induces internalization of TRPA1 in sensory neurons, including neurons lacking TRPV1 expression (i.e. TG neurons from TRPV1 null-mutant mice). In TG neurons cultured from rats, no detectable MO-induced internalization of either

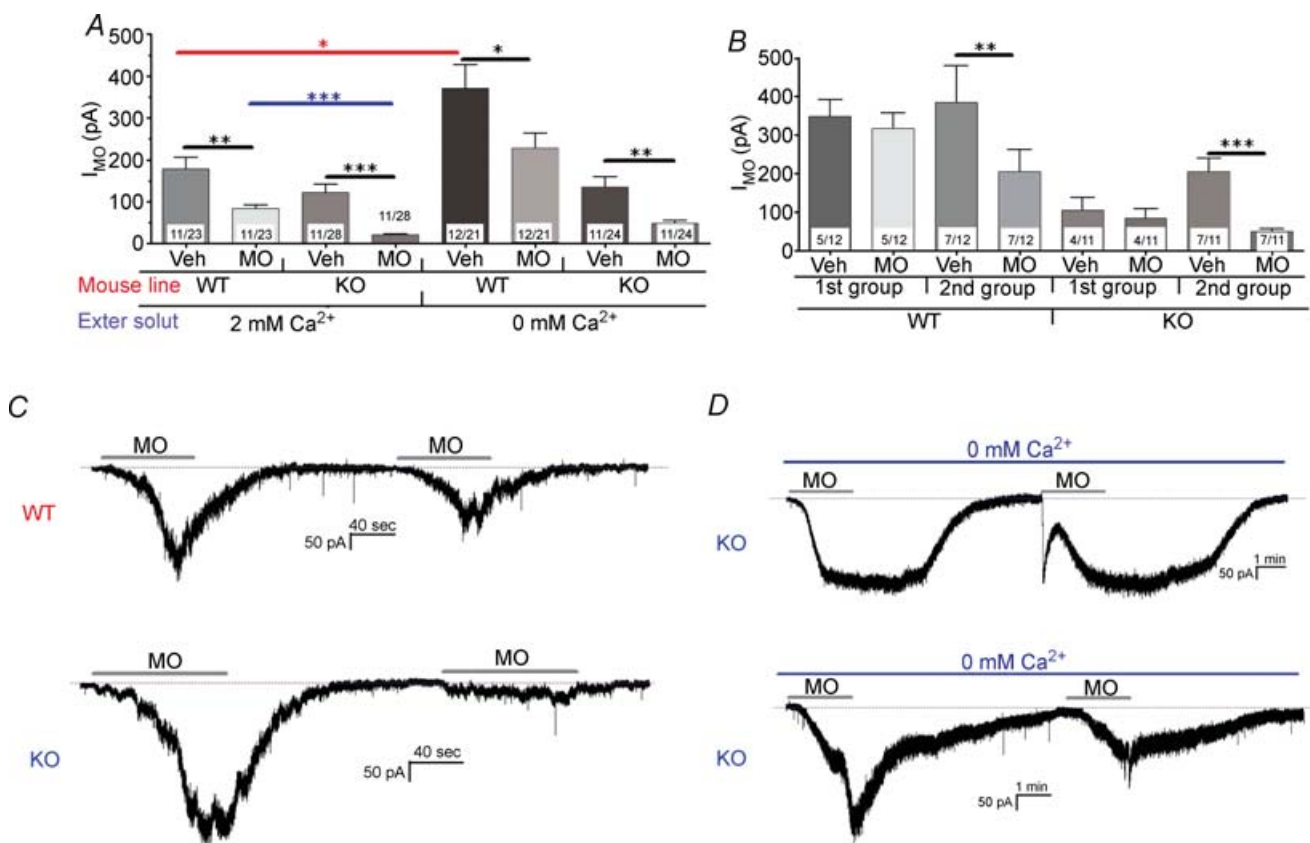


Figure 7. TRPV1 regulates pharmacological desensitization of TRPA1 by MO

Recordings were made of wild-type and TRPV1 KO mouse sensory neurons using perforated patch configuration. A, I_{MO} tachyphylaxis was measured in WT and TRPV1 KO mice. Standard (2 mM Ca^{2+}) and Ca^{2+} -free (0 mM Ca^{2+}) extracellular solutions were used to examine I_{MO} (50 μ M) tachyphylaxis. Significant differences between values of various groups are marked over horizontal bars connecting the bars which represent those groups. Treatments, mouse lines and the amount of Ca^{2+} in the extracellular solution are indicated below the X-axis. Note that the blue horizontal bar refers to statistically significant difference between I_{MO} in the WT versus TRPV1 KO on the second application of MO. The red horizontal bar indicates significant increase in I_{MO} recorded in Ca^{2+} -free solution (0 Ca-ES) versus in standard extracellular solution (SES). B, separate presentation of I_{MO} tachyphylaxis in Ca^{2+} -free extracellular solution for two types of neurons. This putative division into group of neurons can be made for both WT and TRPV1 KO mouse neurons. C, representative traces demonstrate I_{MO} tachyphylaxis in neurons from WT and TRPV1 KO mice. D, two different traces illustrate I_{MO} tachyphylaxis in Ca^{2+} -free media for the first (upper panel) and second (bottom panel) groups of neurons isolated from TRPV1 KO mice.

TRPA1 (Fig. 9A; $-5 \pm 5.8\%$, $n = 3$) or TRPV1 (Fig. 9B) was observed. A similar lack of effect was observed in TG neurons cultured from WT mice, where the application of MO had no effect on the internalization of either TRPA1 (Fig. 9D) or TRPV1 (Fig. 9E; see also Supplemental Fig. 5). However, in sensory neurons from TRPV1 null-mutant mice (Fig. 9D; right lanes of the left panel), MO treatment produced a significant reduction ($24.32 \pm 6.4\%$, $n = 4$) in cell surface levels of TRPA1 (Fig. 9D and E, and Supplemental Fig. 5). Altogether, these observations suggest that TRPV1 inhibits the internalization of TRPA1, and that this event could be correlated with more pronounced desensitization of TRPA1 in TRPV1-lacking neurons (or CHO cells).

Discussion

Several studies in humans and animals have demonstrated that treatment with mustard oil, a TRPA1 agonist, can

lead to functional desensitization of sensory systems (i.e. taste, olfactory or somatosensory) to a variety of noxious and chemical stimuli, including capsaicin (Reeh *et al.* 1986; Patacchini *et al.* 1990; Heyer *et al.* 1991; Brand & Jacquot, 2002; Simons *et al.* 2004). Further, the application of capsaicin, a TRPV1-specific agonist, can heterologously desensitize MO responses (Patacchini *et al.* 1990; Jacquot *et al.* 2005). The TRPA1 channel has important roles in processing sensory information including noxious stimuli in a variety of physiological and pathophysiological conditions (Bandell *et al.* 2004; Bautista *et al.* 2006; Jeske *et al.* 2006; Kwan *et al.* 2006). Therefore, changes in channel activities through two fundamental aspects, sensitization and/or pharmacological desensitization, could contribute to the overall sensitivity of nociceptors and a resulting perception of pain. Accordingly, the aims of the present study were to determine the molecular mechanisms mediating MO- and CAP-evoked pharmacological desensitization of TRPA1 in

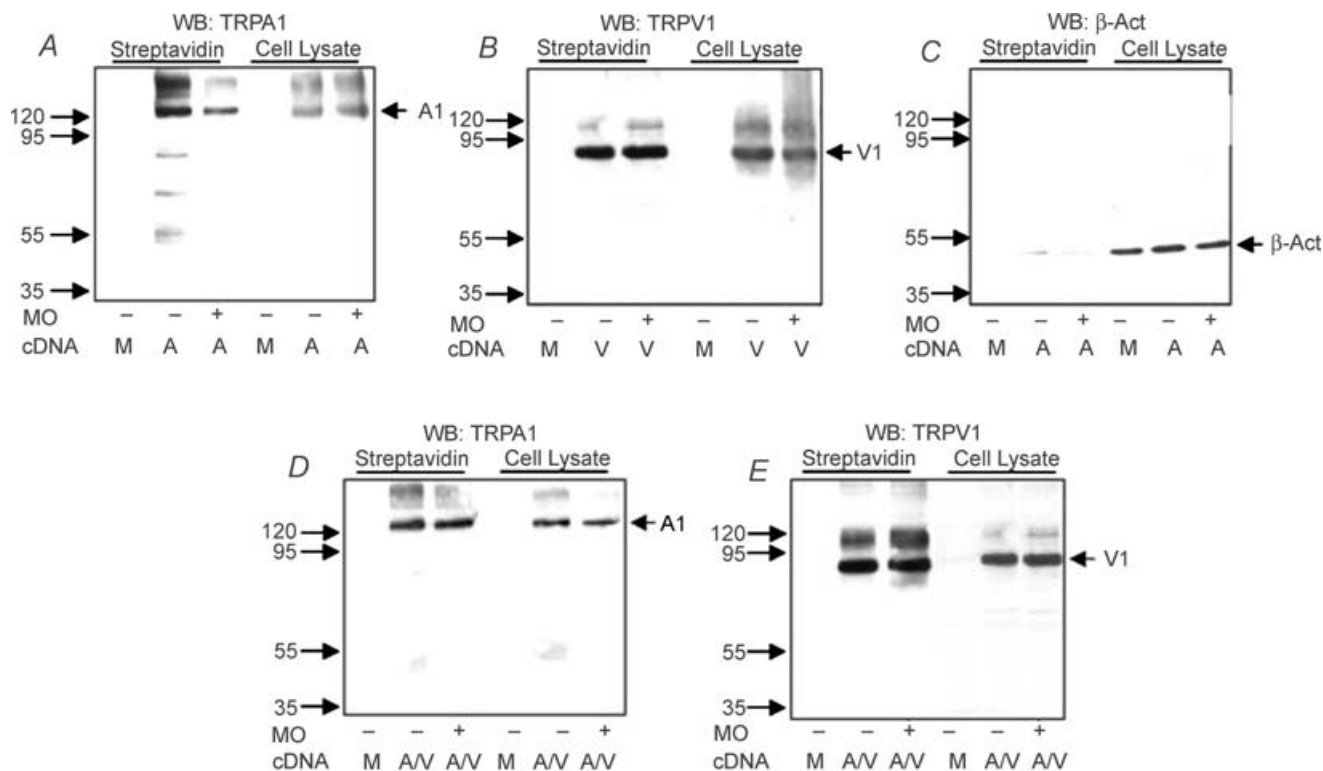


Figure 8. TRPV1 controls MO-evoked TRPA1 internalization in transfected CHO cells

CHO cells were transfected with no cDNA (i.e. mock transfection), TRPA1 (A and C), TRPV1 (B) or TRPA1/TRPV1 (D and E). Transfection constructs (M, mock; V, TRPV1; A, TRPA1; A/V, TRPA1/TRPV1) are indicated as 'cDNA' below blots. Cells were treated for 20 min with MO (50 μ M; treatment is noted below blots). After treatment, cell lysate ('Cell Lysate') and streptavidin-precipitated ('Streptavidin') biotinylated cell surface proteins from separately transfected TRPA1- (A and C) and TRPV1- (B) expressing CHO cells or cotransfected TRPA1/TRPV1-expressing CHO cells (D and E) were isolated and immunoblotted with appropriate antibodies noted above blots. Membranes illustrated in A were stripped and re-probed for β -actin (C) as a negative control for surface biotinylation of proteins. Cell lysate and TRPV1 (for D and E only) served as an internal normalization control in these experiments. Molecular weight markers are shown on left side of each blot. Bands corresponding to TRPA1, TRPV1 and β -actin are marked by arrows on right sides of the blots.

sensory neurons, as well as the mechanisms responsible for MO-evoked desensitization of TRPV1. The data presented above suggest that: (1) consistent with reports showing involvement of calcineurin in the pharmacological desensitization of TRPV1 (Docherty *et al.* 1996; Piper *et al.* 1999; Mohapatra & Nau, 2005), MO-induced inhibition of TRPV1 recruits the Ca^{2+} -dependent calcineurin pathway; (2) CAP-induced pharmacological desensitization of TRPA1 is Ca^{2+} dependent and involves PIP_2 depletion; (3) MO-evoked TRPA1 desensitization in sensory neurons is not influenced by extracellular Ca^{2+} ; (4) CAP can activate the PLC pathway leading to subsequent PIP_2 depletion in both sensory neurons and TRPV1-expressing CHO cells, while MO is capable of activating PLC in TRPA1-overexpressing CHO cells, but not in sensory neurons; (5) the magnitude of TRPA1 desensitization is regulated by the presence of TRPV1, which acts to preserve TRPA1 activity; and (6) the apparent

MO-induced internalization of TRPA1 occurs only in the absence of TRPV1.

The mechanisms for acute desensitization has been the topic of most studies on desensitization of TRP channels (Docherty *et al.* 1996; Nilius *et al.* 2005, 2006; Rohacs *et al.* 2005; Zhu *et al.* 2005). In agreement with other reports (Story *et al.* 2003; Bandell *et al.* 2004; Jordt *et al.* 2004; Nagata *et al.* 2005; Zurborg *et al.* 2007), we found that in TRPA1-expressing CHO cells, I_{MO} is acutely desensitized in the presence of Ca^{2+} (Fig. 2C–F and Supplemental Fig. 1). It appears that the removal of extracellular Ca^{2+} affects acute desensitization of I_{MO} in TRPA1-expressing cells (Nagata *et al.* 2005). However, in sensory neurons, the acute desensitization of I_{MO} within 2 min of drug application was not apparent and was not affected by Ca^{2+} (Figs 1C and D, and 3C–E). Moreover, MO was capable of inducing PLC activity in TRPA1-expressing CHO cells but not sensory neurons (Figs 5 and 6). This difference

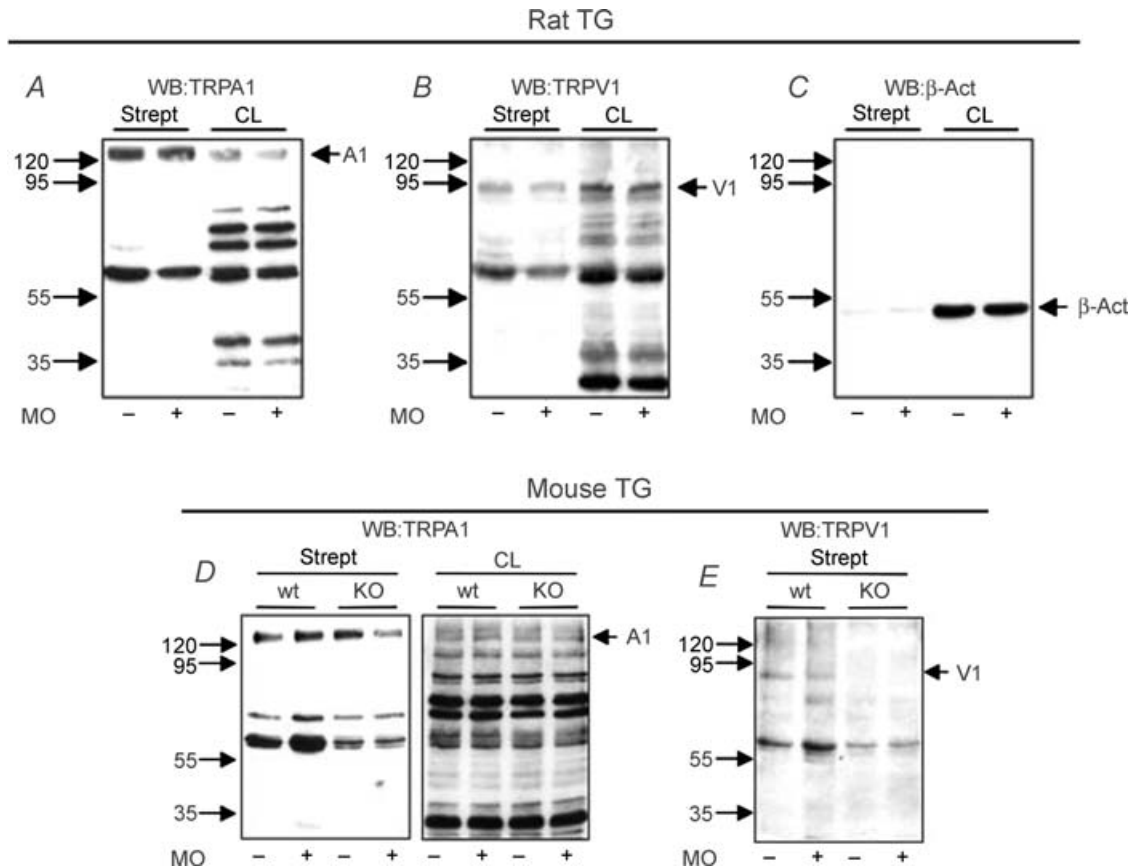


Figure 9. TRPV1 controls MO-evoked TRPA1 internalization in sensory neurons

Trigeminal sensory neurons isolated from rat (A–C), wild-type and TRPV1 KO mouse (D and E) were treated for 20 min with MO (50 μM ; treatment is noted below blots). After treatment, cell lysate ('CL') and streptavidin-precipitated ('Strept') biotinylated cell surface proteins were isolated and immunoblotted with appropriate antibodies noted above blots. Membranes illustrated in A were stripped and re-probed with β -actin (C) to demonstrate that biotinylation was restricted to surface proteins. Cell lysate and TRPV1 (for A and D only) served as an internal normalization control in these experiments. Molecular weight markers are shown on left side of each blot. Bands corresponding to TRPA1, TRPV1 and β -actin are marked with arrows on right sides of the blots.

could contribute to the differences in acute desensitization between sensory neurons and TRPA1 over-expressing CHO cells.

Acute and pharmacological desensitization could share common pathways in their mechanisms. However, acute and pharmacological desensitization differ from each other in one major aspect. The mechanisms responsible for gradual loss of channel activity in the continued presence of specific agonists underlie acute desensitization. In contrast, repeated agonist application leading to pharmacological desensitization can be divided into two separate events which are loss of activity (i.e. desensitization) after drug treatment (not necessarily from a homologous agonist) and recovery from lost activity (i.e. re-sensitization), which is triggered by the same drug causing desensitization. Moreover, the balance between desensitization and re-sensitization rates depends on the time after drug treatment (Jung *et al.* 2004). Therefore, the modulation of either event could critically contribute to the measured outcome of pharmacological desensitization. Thus, pharmacological desensitization of TRPV1 could be prevented by either blockage of calcineurin-induced dephosphorylation (Mohapatra & Nau, 2005) or by activation of protein kinase A (Bhave *et al.* 2002; Mohapatra & Nau, 2005), PKC (Mandadi *et al.* 2004) or Ca^{2+} -dependent calmodulin kinase (Jung *et al.* 2004; Rosenbaum *et al.* 2004). From this perspective, the effects of PIP_2 on CAP-mediated inhibition of TRPA1 could be interpreted as contributing to either desensitization or re-sensitization. Intracellular ATP is one of the major components contributing to re-sensitization of TRP channels (Bhave *et al.* 2002; Liu & Qin, 2005; Liu *et al.* 2005; Mohapatra & Nau, 2005). Therefore, to minimize the contribution of re-sensitization pathways in our mechanistic studies focusing on pharmacological desensitization (Fig. 4 and Supplemental Fig. 4), we performed whole-cell recording without ATP and GTP in the pipette solution, and did not employ PI4-kinase activity blockade (by $10 \mu\text{M}$ wortmanin). Our results provide two major lines of evidence suggesting that CAP-evoked PIP_2 depletion is one of the major pathways responsible for desensitization of TRPA1. First, previous studies have demonstrated that certain TRP channels can be activated by PIP_2 and/or their activities run-down when recordings are made in whole-cell configuration (Liu & Qin, 2005; Rohacs *et al.* 2005; Nilius *et al.* 2006). We have not detected activation of TRPA1 with the delivery of PIP_2 (data not shown). The magnitude of I_{MO} run-down was minimal after 5 min of establishing whole-cell configuration (Fig. 4A; for perforated patch, $I_{\text{MO}} = -231 \pm 31.9 \text{ pA}$, $n = 14$ versus for whole-cell after 5 min, $I_{\text{MO}} = -186.5 \pm 25.9 \text{ pA}$, $n = 12$, t test n.s.; Fig. 4A). However, the run-down was significant and apparent after 15 min of establishing whole-cell configuration (see numerical data in Results; Fig. 4B).

In addition, the intracellular delivery of exogenous PIP_2 to sensory neurons partially reversed this reduction in I_{MO} magnitude (Fig. 4B). Second, the application of CAP induced a prompt depletion, rather than replenishment, of PIP_2 in sensory neurons and TRPV1-expressing cells (Figs 5 and 6). Altogether, these data support the hypothesis that CAP-evoked Ca^{2+} -dependent depletion of PIP_2 in sensory neurons leads to a down-regulation of TRPA1 channel activity and subsequent desensitization.

The desensitization of a TRP channel could involve several independent pathways. Thus, TRPV1 desensitization is regulated by calcineurin-triggered dephosphorylation (Jung *et al.* 2004; Mohapatra & Nau, 2005) and/or PIP_2 depletion (Liu *et al.* 2005). TRPM8 desensitization is regulated by Ca^{2+} -dependent, PKC-directed phosphorylation (Abe *et al.* 2006) and/or PIP_2 depletion (Rohacs *et al.* 2005). The results from the present study indicate that, at least in sensory neurons, calcineurin activation is sufficient to pharmacologically desensitize TRPV1 by CAP or MO (Supplemental Fig. 3). These results are consistent with previous reports (Jung *et al.* 2004; Mohapatra & Nau, 2005). However, in TRPV1-overexpressing cell lines, the depletion/replenishment of PIP_2 could contribute to TRPV1 pharmacological desensitization (Liu *et al.* 2005). These results do not exclude a possible dual regulation of TRPV1 by phosphorylation and PIP_2 biosynthesis (Chuang *et al.* 2001; Liu *et al.* 2005). Moreover, cooperation between PIP_2 biosynthesis and channel phosphorylation, a mechanism proposed previously for certain voltage-gated calcium channels (Wu *et al.* 2002), could play a role in regulation of TRPV1 activities in sensory neurons.

Unlike TRPV1, the pharmacological desensitization of TRPA1 can recruit two different pathways via CAP-evoked Ca^{2+} -dependent and MO-evoked Ca^{2+} -independent inhibition of TRPA1 responses (Fig. 3). Based upon the present findings, we propose that the rapid, substantial and prolonged elevation in $[\text{Ca}^{2+}]_i$ in sensory neurons induced by CAP, but not MO, triggers PIP_2 depletion that, in turn, attenuates TRPA1 activities. This CAP-evoked hydrolysis of PIP_2 was observed in almost all treated sensory neurons (Fig. 5). Since TRPA1 is a Ca^{2+} -permeable channel (Story *et al.* 2003; Jordt *et al.* 2004), it is possible that the application of MO could result in PIP_2 hydrolysis. However, it appears that either the amount of $[\text{Ca}^{2+}]_i$, the speed of Ca^{2+} influx, the duration in maintenance of $[\text{Ca}^{2+}]_i$, or some combination of these events generated by MO treatment is sufficient to trigger PIP_2 depletion only in ~15–20% of MO-sensitive sensory neurons (Fig. 5). In contrast, MO was able to induce PIP_2 hydrolysis in TRPA1-overexpressing CHO cells, and these differences are possibly due to evoking a sufficient accumulation of $[\text{Ca}^{2+}]_i$ to activate PLC in CHO cells but not in sensory neurons (Fig. 5 versus

Fig. 6). Alternatively, it is possible that CHO cells may have a greater concentration of Ca^{2+} -dependent PLC isoforms. Despite the finding that MO was not effective in hydrolysing PIP_2 in neurons, the MO-generated elevation in $[\text{Ca}^{2+}]_i$ was sufficient to activate the Ca^{2+} -dependent phosphatase calcineurin, which is responsible for TRPV1 desensitization (Supplemental Fig. 3A). A plausible explanation of this mechanism is that calcineurin is more sensitive to changes in $[\text{Ca}^{2+}]_i$ than PLC or is located nearby the pore region where transient $[\text{Ca}^{2+}]_i$ levels might be greater. This could be due to the known high affinity interaction between the calcineurin–calmodulin complex and Ca^{2+} (Hashimoto *et al.* 1988).

A suprathreshold elevation in $[\text{Ca}^{2+}]_i$ is necessary to achieve acute as well as pharmacological desensitization of the TRP channels (Koplas *et al.* 1997; Liu *et al.* 2005; Nilius *et al.* 2005, 2006; Rohacs *et al.* 2005; Abe *et al.* 2006). In this respect, Ca^{2+} -independent desensitization of TRPA1 by MO is unique. The precise molecular mechanisms for this Ca^{2+} -independent inhibition of TRPA1 activities are unknown as of yet. However, there is a correlation between the magnitude of I_{MO} and the rate of its tachyphylaxis (Figs 1A and 3B). Moreover, since CAP could not activate this pathway (Fig. 3A), Ca^{2+} -independent desensitization of the TRPA1 channel may require TRPA1 activation by its own agonist. Recent studies on mechanisms of TRPA1 activation by MO demonstrated that covalent modification of reactive cysteines by MO within TRPA1 can cause channel activation (Hinman *et al.* 2006; Macpherson *et al.* 2007). These covalent modifications are slowly reversible, and the reversibility depends on external Ca^{2+} (Macpherson *et al.* 2007). Interestingly, I_{MO} can be maintained above baseline for at least 3–8 min after washout of MO in the Ca^{2+} -free solution (Fig. 3C–E; Macpherson *et al.* 2007). From this perspective, one could hypothesize that the covalent modification of TRPA1 channels may persist for longer periods based upon the magnitude of the initial I_{MO} , and this model might explain the observation that repeated application of MO generates pharmacological desensitization of the TRPA1 channel in the absence of extracellular Ca^{2+} . However, additional unknown mechanisms for Ca^{2+} -independent I_{MO} tachyphylaxis in sensory neurons cannot be excluded. The presence of an unknown mechanism is suspected because in some sensory neurons, I_{MO} tachyphylaxis did not occur even after detection of initial TRPA1 activation or under conditions where MO was reapplied after the initial I_{MO} declined to baseline (i.e. covalent modification could have been substantially reversed; Hinman *et al.* 2006). Moreover, other preliminary data suggest that another TRPA1 agonist, the cannabinoid WIN55 212 (Jeske *et al.* 2006), is also able to desensitize I_{MO} and I_{WIN} in Ca^{2+} -free extracellular media (A. N. Akopian and K. M. Hargreaves, unpublished observations). However, cannabinoids cannot produce covalent modification

of cysteine residues since these compounds do not contain thiol adducts (Hinman *et al.* 2006). Together, these observations support the possibility of additional mechanisms for Ca^{2+} -independent I_{MO} tachyphylaxis in sensory neurons.

Our studies on I_{MO} tachyphylaxis in TRPA1- or TRPA1/TRPV1-expressing CHO cells demonstrated that I_{MO} underwent substantially greater tachyphylaxis in the absence of TRPV1 (Fig. 2A). We next confirmed these effects in sensory neurons (Fig. 7A), since this unusual response pattern of I_{MO} could be attributed to a cell-specific biochemistry of CHO cells. One of the feasible explanations for this event is that TRPV1 stabilizes the membrane surface expression of TRPA1 (Figs 8 and 9 and Supplemental Fig. 5). It is possible that the relative proportions in cell surface expression of TRPA1 and TRPV1 vary from neuron to neuron, or in microdomains within the same cell. Therefore, it could be suggested that TRPA1 internalization could occur in at least a subset of sensory neurons. The cell surface biotinylation approach, which detects internalization of the receptor on a cell population basis, may not be suitable for detecting translocation of TRPA1 in a relatively small neuronal subpopulation. Under normal physiological conditions, TRPV1-mediated regulation of TRPA1 internalization might not be detected in sensory neurons due to the high level of coexpression of these TRP channels. However, changes in relative TRPV1 and TRPA1 cell surface expression that occur during tissue damage or nerve injury (Nagy *et al.* 2004; Obata *et al.* 2004, 2005) could influence regulatory mechanisms of TRPA1 activities. According to this hypothesis, both TRPA1 and TRPV1 contribute to the actions of inflammatory mediators on sensory neurons (Chuang *et al.* 2001; Bandell *et al.* 2004; Bautista *et al.* 2006; Kwan *et al.* 2006), and it is possible that the relative balance of these activities may be altered during certain pathophysiological conditions. Therefore, an important objective for future studies will be to determine whether TRPV1 regulates TRPA1 activities in a variety of tissue damage and nerve injury pain models.

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Supplemental material

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